

Finite Element Analysis in Biomedical Engineering

376-1712-00

Assignment 2

Max 2 students per solution!

Send the report and your Matlab/Python routines to ***bhelgason@ethz.ch*** no later than **20 June, 2026**. You are allowed to use AI tools to assist you in the coding process. Please report which AI tools you used and describe in your report the prompting approach you took.

Task 1 (20%): Program an FE solver in Matlab or Python that can solve general 3D structural problems with 4-node tetrahedral finite elements.

To guide your coding you can use a 2D version of the solver uploaded to the Moodle page.

The function has to have the following form:

`[u,epsi,epsil,epsil3,sigma,sigma1,sigma3]=solid3D(nodes,elements,loadl,bcs,plot_scale);`

The variables *nodes*, *elements*, *loadl*, *bcs* are read from an input file. Example of such input files can be downloaded from the course Moodle page (*test_3D.m* and *test4_3D.m*). The *plot_scale* variable is for controlling the visualization of the deformed structures.

The input file has the form:

nodes = [node nr, x, y, z]
elements = [elem nr, node nr i, node nr j, node nr k, node nr l, E, ν]
loadl = [node nr, Fx, Fy, Fz] (concentrated forces)
bcs = [node nr, ux, uy, uz]

The function needs to output:

u: nodal degrees of freedom solution
epsi: Vector with the normal and shear strains for all the elements
sigma: Vector with the normal and shear stresses for all the elements
epsil: Vector with the 1st principal strains for all the elements
sigma1: Vector with 1st principal stresses for all the elements
epsil3: Vector with the 3rd principal strains for all the elements
sigma3: Vector with 3rd principal stresses for all the elements

Task 2 (0%). Segment, mesh and map material properties to a FEM of a human femur using MITK-GEM. Use the input data which can be found on the Moodle page. Use the same calibration as we used in the MITK-GEM tutorial.

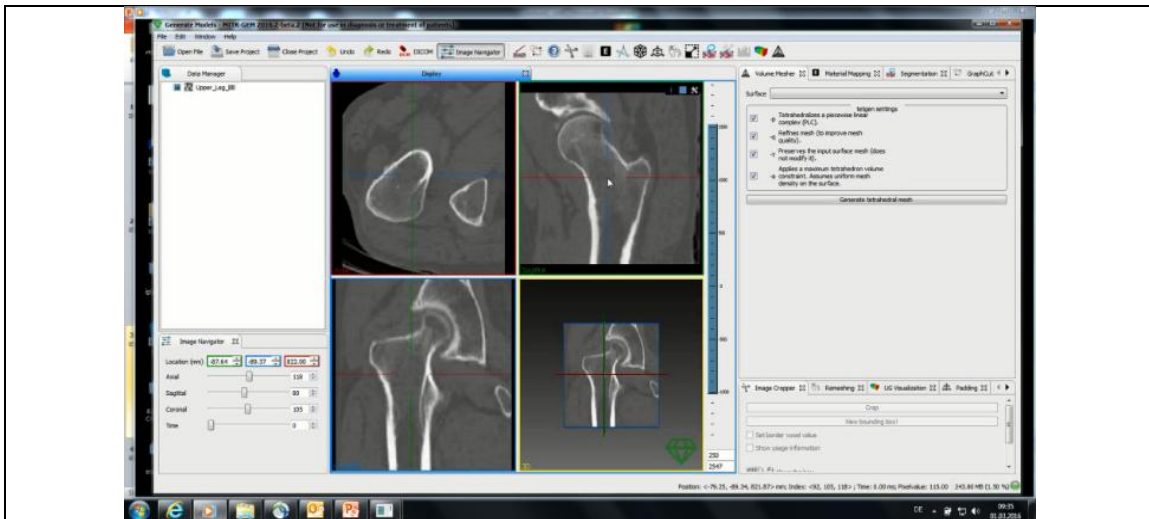


Fig. 1: Segmented femur.

Task 3 – sideways fall loading configuration (20%): Use your 3D solver to analyze the FEM of the femur that you segmented in Task 2. The function *WriteFemurInputFile_rev2.m* on the Moodle page writes the input file for you using your MITK-GEM output file. Be aware of that the model is using a displacement of 1 mm at the greater trochanter so that you need to implement the second BC method from the lecture notes to address that. Write a MATLAB function that draws:

- the input (see figure 1a and 1b)
- the output (e.g., σ_1 , σ_3 , or deformations, see figure 1c)

Useful built-in MATLAB functions: *tetramesh*,

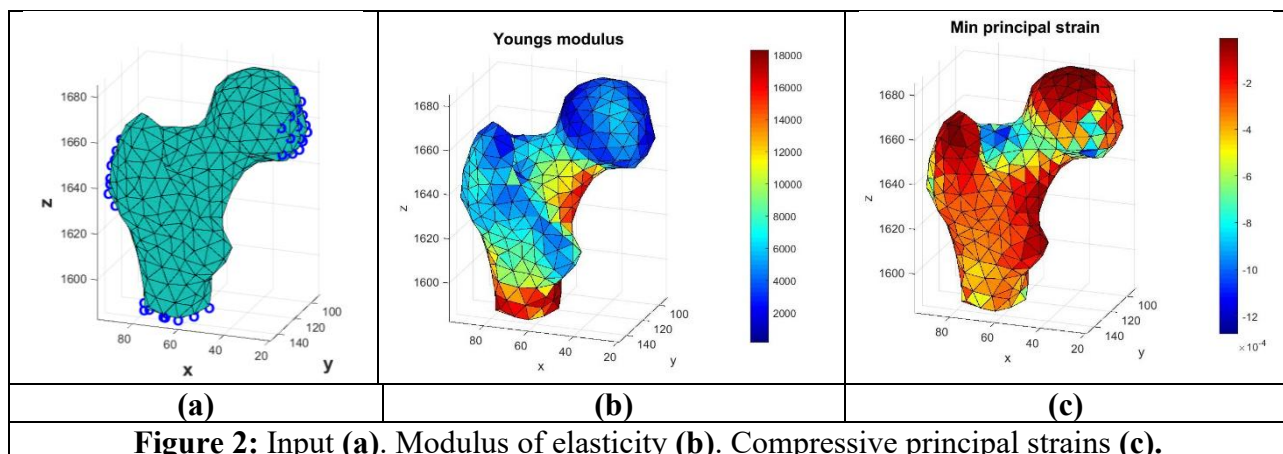
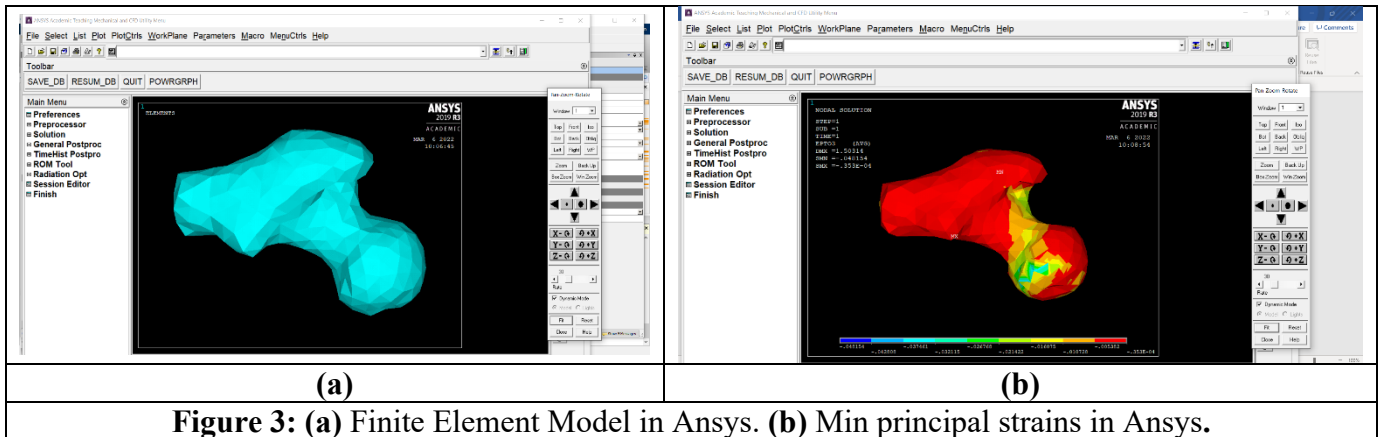


Figure 2: Input (a). Modulus of elasticity (b). Compressive principal strains (c).

Task 4 – Single leg stance loading configuration (20%): Use your 3D solver to analyze your femur model under single leg stance loading. Constrain the distal end in all directions and apply a force on the femoral head (under a meaningful physiological angle) corresponding to the sum of the body weight of your team.

Task 5 (20%): Repeat the analysis in Task 3 using Ansys Workbench. The *WriteAnsysInput_rev2.m* should generate a file that is named *RunAnsysModelB.inp*, which is the file you launch in Ansys to run the analysis and plot strains.



Tasks 6 (20%)

Write a 4-5 page report where your solution to the assignment is described. Report e.g.:

- Did you use AI tools to help you with your coding? Describe which tools and your prompt strategy.
- What was the outcome of your 1- and 4- element tests using your code?
- And did you get the same results for the femur using your FE solver (Task 3) and Ansys Workbench (Task 5)? If there is a difference, what could be the explanation for that?
- What are the main differences in principal strain patterns you get between Task 3 (sideways fall loading) and Task 4 (single leg stance loading).